Anh Thuan Tran

(571) 373 6168 | atran92@gmu.edu | linkedin.com/in/atran1999 | github.com/anhthuan1999 | Google Scholar

Summary

Ph.D. Student in Computer Science with research experience on Computer Vision, Deep Learning, 3D Gaussian Splatting, 3D vision, 3D point cloud understanding, and vision–language models.

Education

George Mason University — Virginia, USA
Ph.D in Computer Science | GPA: 3.89 — Sep 2024 – Exp. May 2029

- **Research Interest**: 3D Vision, 3D Gaussian Splatting, 3DGS SLAM, Dynamic 3D Reconstruction, Vision Language
- **Advisor**: Prof. Jana Kosecka

Pukyong National University — Busan, South Korea
M.Eng. in Artificial Intelligence Convergence | GPA: 3.89 — Mar 2022 – Feb 2024

Vietnam National University — Ho Chi Minh City, Vietnam
B.Ec. in E-commerce | GPA: 3.42 (Thesis track) — Sep 2017 – Jul 2021

Publications

[1] **Anh Thuan Tran** and Jana Kosecka. VarSplat: Uncertainty-aware 3D Gaussian Splatting for Robust RGB-D SLAM. *Under Review*, 2025. [PDF].

[2] **Anh Thuan Tran** and Jana Kosecka. PointSplat: Unified Image-Agnostic Gaussian Pruning for 3D Transformer Refinement. *Under Review*, 2025. [PDF].

[3] **Anh-Thuan Tran**, Hoanh-Su Le, Suk-Hwan Lee, and Ki-Ryong Kwon. PointCT: Point Central Transformer Network for Weakly-Supervised Point Cloud Semantic Segmentation. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 3556–3565, 2024. [PDF | Code].

Research Experience

Artificial Intelligence and Database Lab — GMU, USA
Graduate Research Assistant — Sep 2024 – Present

- **VarSplat** (SLAM & Uncertainty): Built an uncertainty-aware RGB-D 3DGS-SLAM system, rendering per-pixel uncertainty via the law of total variance to improve long-sequence robustness.
- **PointSplat** (Pruning): Developed image-agnostic pruning and refinement for 3D Gaussians in medium-scale indoor scenes with Point Transformer network.
- Dynamic 3DGS (Ongoing): Developing continual learning method to update 3D Gaussian maps in changing environments while excluding distractors such as moving people.
- Open-Vocabulary 3DGS: Implemented vision-language baseline by integrating LSeg features or CLIP with DINOv2 and SAM into 3D Gaussian attributes.

Multimedia Signal Processing Lab PKNU, South Korea
Graduate Research Assistant Mar 2022 – Feb 2024

- **PointCT**: Proposed a point-based Transformer network for weakly-supervised point cloud segmentation.
- Developed central-based attention block to integrate global features across neighborhoods, and improve unlabeled point representations.
- Achieved competitive performance on the S3DIS, ScanNet with 0.1% labeled points, STPLS3D with 0.01% labeled points.

Work Experience

Newgen I&S Busan, South Korea
AI Researcher Jan 2024 – Jul 2024

- Developed offline chatbots for marine industry using LangChain and Ollama to process bilingual (English–Korean) PDFs.
- Processed text with FAISS, updated PostgreSQL, and LLM responses with top-k retrieval.
- Deployed system on server with WebSocket for real-time communication.

Scholarships

- Graduate Teaching Assistant, George Mason University (Sep 2024 - Present)
- Brain Korea 21, Graduate Student Research Scholarship (Artificial Intelligence Convergence Research Group), Pukyong National University (Mar 2022 - Feb 2024)
- Full-time Scholarship For Graduate Student, Pukyong National University (Mar 2022 - Feb 2024)
- Graduate student scholarship, Pukyong National University (Sep 2022 - Feb 2023)

Skills

- **Programming:** Python — Proficiency: Advanced
- **ML & 3D Vision:** 3D Gaussian Splatting, NeRFs, 3D Vision, Point Clouds, 3D Point Transformers, 2D–3D Distillation, Vision–Language Models, Open-Vocabulary Understanding
- **Frameworks & Libraries:** PyTorch, PointOps, NumPy, OpenCV, CLIP, DINO, NeRFStudio, Differentiable Gaussian Rasterization with Depth, Pose, and Variance.
- **Tools & Systems:** Ubuntu, Slurm, Blender, LangChain, Ollama, PostgreSQL, FAISS, RAG.

Awards

- Outstanding International Student Award - Pukyong National University (2024)
- Excellent achievement in undergraduate research – VNU-HCM (2022)
- Third Prize Young Scientist – University of Economics and Law, VNU-HCM (2018, 2020)